

Program : Diploma in Information Technology	
Course Code : 5269	Course Title: Server Administration Lab
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To apply the principles, practices and goals of server administration.
- To provide practical experience on administrating networking, FTP and print servers
- To configure samba and NFS file servers.

Course Pre-requisites:

Topic	Course Code	Course Title	Semester
Basics of networking		Fundamentals of network administration	3
Basic knowledge of System administration		System administration	4

Course Outcomes:

On completion of the course, students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop basic concept of server administration and SSH software packages	10	Applying
CO2	Make use of the basic concept in administrating networking, IP addressing and network interfaces	10	Applying
CO3	Apply basic concepts of print server, web server and FTP server	11	Applying
CO4	Apply basic concepts of samba and NFS file server	11	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3		3				
CO3	3		3				
CO4		3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Develop basic concept of server administration and SSH software packages		
M1.01	Make use ofSSH command, log into another computer	2	Applying
M1.02	Identify remote execution with SSH command	2	Applying
M1.03	Make use of SSH tools in client server software packages	2	Applying
M1.04	Experiment with Configuring SSH	2	Applying
M 1.05	Apply SSH for remote login and remote execution	2	Applying
CO2	Make use of the basic concept in administrating networking, IP addressing and network interfaces		
M2.01	Experiment with Configuring Linux as a DHCP server, DNS, Router	2	Applying
M2.02	Build network proxy connection.	1	Applying
M2.03	Experiment with Configuring network interfaces	1	Applying
M2.04	Make use of command to check the routes in local network	2	Applying
M2.05	Identify the network manager to turn off and disable network.	2	Applying
M2.06	Experiment with Configuring IP address on switch	2	Applying
	Lab Exam – I	1.5	
CO3	Apply basic concepts of print server, web server and FTP server		

M3.01	Utilize basic command in print server, web server and FTP server	2	Applying
M3.02	Identify remote administration of system	2	Applying
M3.03	Experiment with Configuring Apache web server to use key.	2	Applying
M3.04	Build HTTPS connection to web server	3	Applying
M3.05	Experiment with Configuring FTP daemon package	2	Applying
CO4	Apply basic concepts of samba and NFS file server		
M4.01	Experiment with Configuring samba and samba-client packages	2	Applying
M4.02	Apply Start and enable the smb and nmb services	2	Applying
M4.03	Develop the samba server workgroup	2	Applying
M4.04	Experiment with the installation of packages needed to configure an NFS service.	3	Applying
M4.05	Make use of the package that provides the NFS server software	2	Applying
	Lab Exam – II	1.5	

***Suggested Open-ended Experiments:

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups. Open ended experiments should include the SSH software packages, administrating networking, IP addressing)

Example:

1. Configure your apache web server to use key and self-signed certificate to serve secure (HTTPS) content.
2. Use the desktop to check that Network Manager has successfully started your network interface to the network. If it has not, then try to start your network interface.
3. Try to contact google.com from the command line in a way that ensures that DNS is working properly.

Text / Reference:

T/R	Book Title / Author
T1	Michael Jang, Alessandro Orsaria “RHCSA/RHCE Red Hat Certification Study Guide”, Seventh Edition
R1	Christopher Negus “Linux Bible” the comprehensive tutorial resources, Ninth Edition

Online Resources:

Sl.No	Website Link
1	https://www.tuxcademy.org/download/en/adm1/adm1-en-manual.pdf
2	https://www.tecmint.com/20-linux-yum-yellowdog-updater-modified-commands-for-package-mangement/
3	https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/6/html/deployment_guide/s2-users-cl-tools